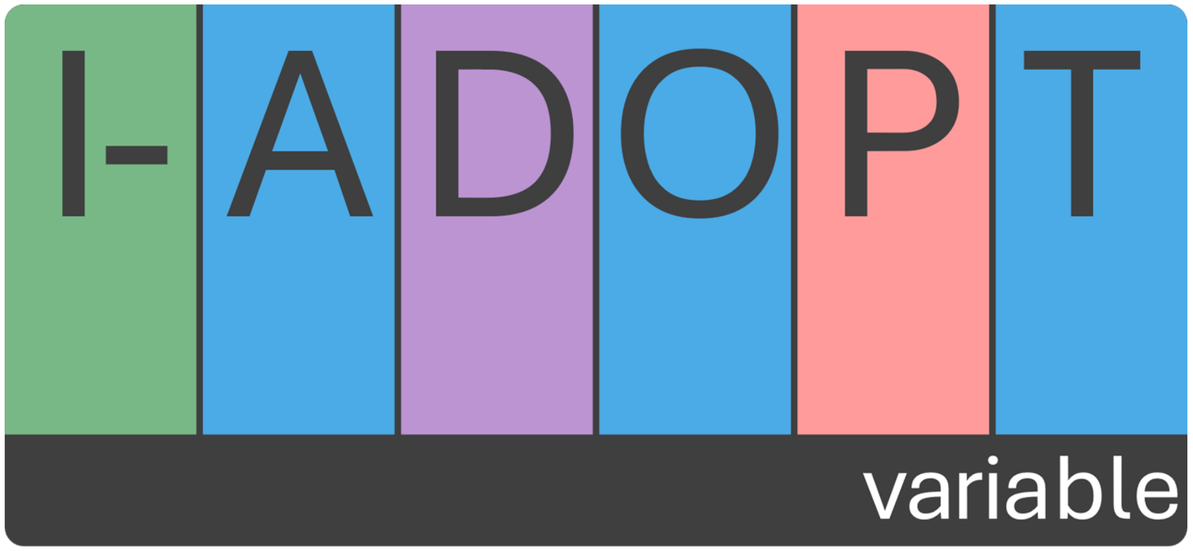
Decomposing Ocean Variables using





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I-ADOPT History

RESEARCH DATA ALLIANCE
I-ADOPT WG

"InteroperAble Descriptions of Observable Property Terminologies"

Chairs:
Barbara Magagna,
GO FAIR Foundation, NL
Gwenaëlle Moncoiffé
BODC, GB
Anusuryia Devaraju
CSIRO, AU
Maria Stoica
University of Colorado, US
Core group:
Sirko Schindler
German Aerospace Center, DE
Alison Pamment
Centre for Environmental Data Analysis, UK

- RDA WG 2019-2021 -> I-ADOPT recommendations (DOI: 10.15497/RDA00071)
- RDA endorsement 2022
- 2022-2024 I-ADOPT vocabularies
- Autumn 2024 Two I-ADOPT Challenges
- Spring 2025 Two I-ADOPT Workshops
- I-ADOPT NFDI4Earth Project (July-November 2025)
- FAIR2Adapt and I-ADOPT (2025-2027)
- CDIF4EOSC (June 2026-May 2029)

The crucial role of metadata

Variables are used to describe which properties are observed & represented in data

However, variables often are:

- Poorly described using free text
- Hard to understand, compare and reuse (especially by machines)

I-ADOPT addresses this challenge:

- Adds machine-readable, standardized variable descriptions
- Makes your data findable, accessible, interoperable, and reusable
- Increases citations and proper reuse of your data

Interoperability Framework

I-ADOPT provides a **FAIR** and systematic **approach** for the description of variables by

- enabling **crosswalks** to a common variable representation across vocabularies
- adding rich human & **machine-actionable** descriptions
- using **qualified**, language independent, **references** using object properties
- specifying the role a concept plays within the variable description
- with **no need to change existing terms and structures**

I-ADOPT variable

Is about the **WHAT**

It does NOT include

- HOW – protocol, instrument
- WHEN – measurement time
- WHERE – specific geographic location
- Value representation – unit (for quantitative properties) or controlled lists (for qualitative properties)

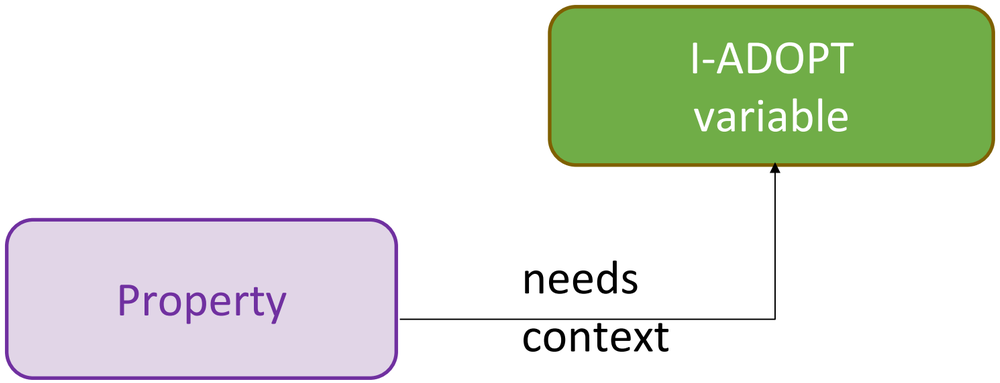
... of the measurement/observation/simulation/calculation

In this way it can be embedded in other standards like SOSA/OMS via the relationship: **sosa:observedProperty** to enrich **sosa:Property** with more consistent and semantic descriptions

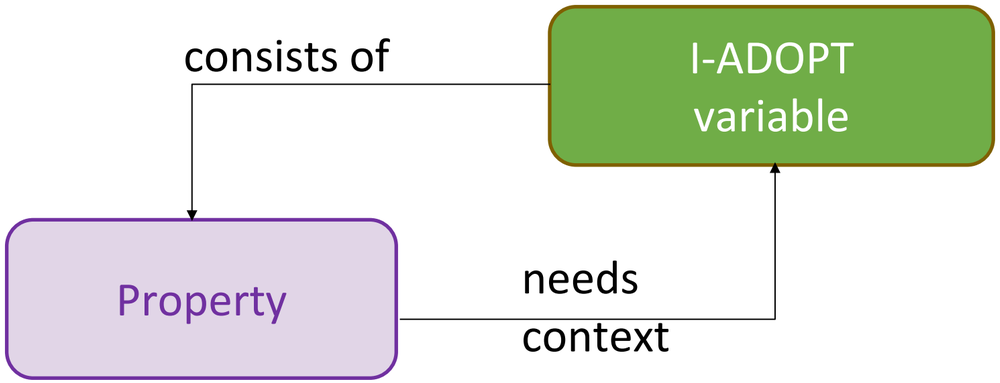
Data description is about a property we observe

Property

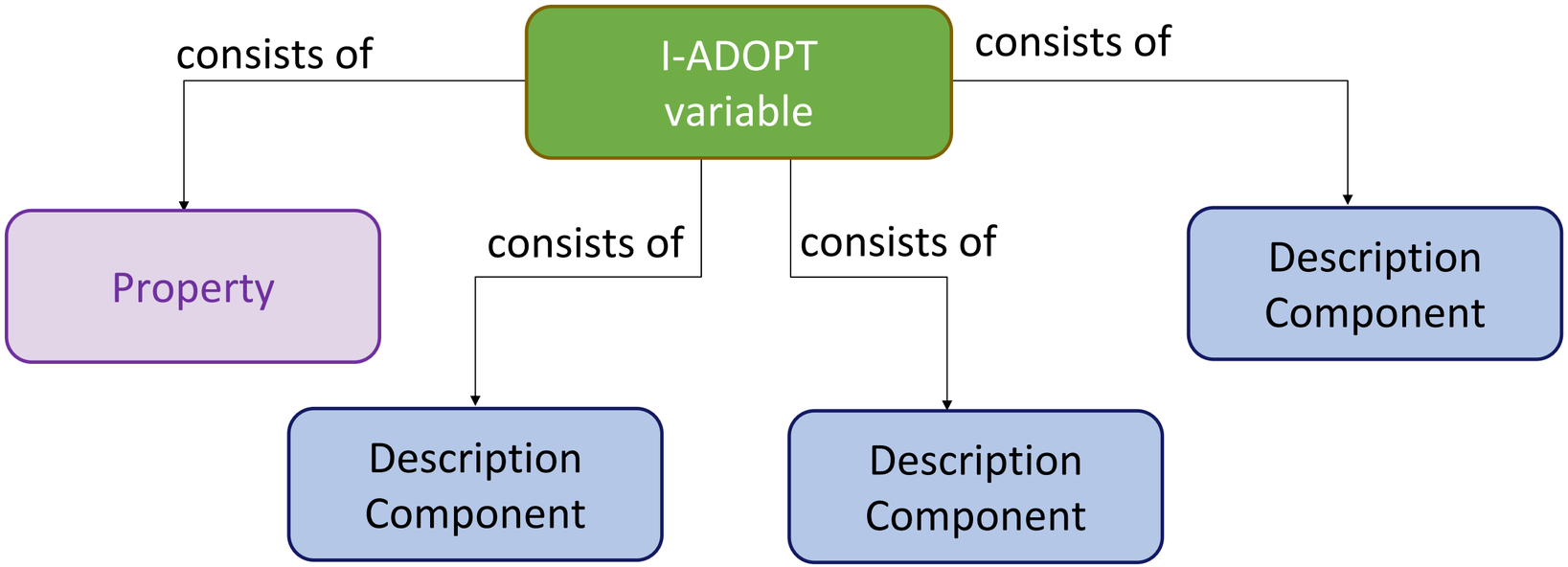
Data description is about a property we observe



Data description is about a property we observe

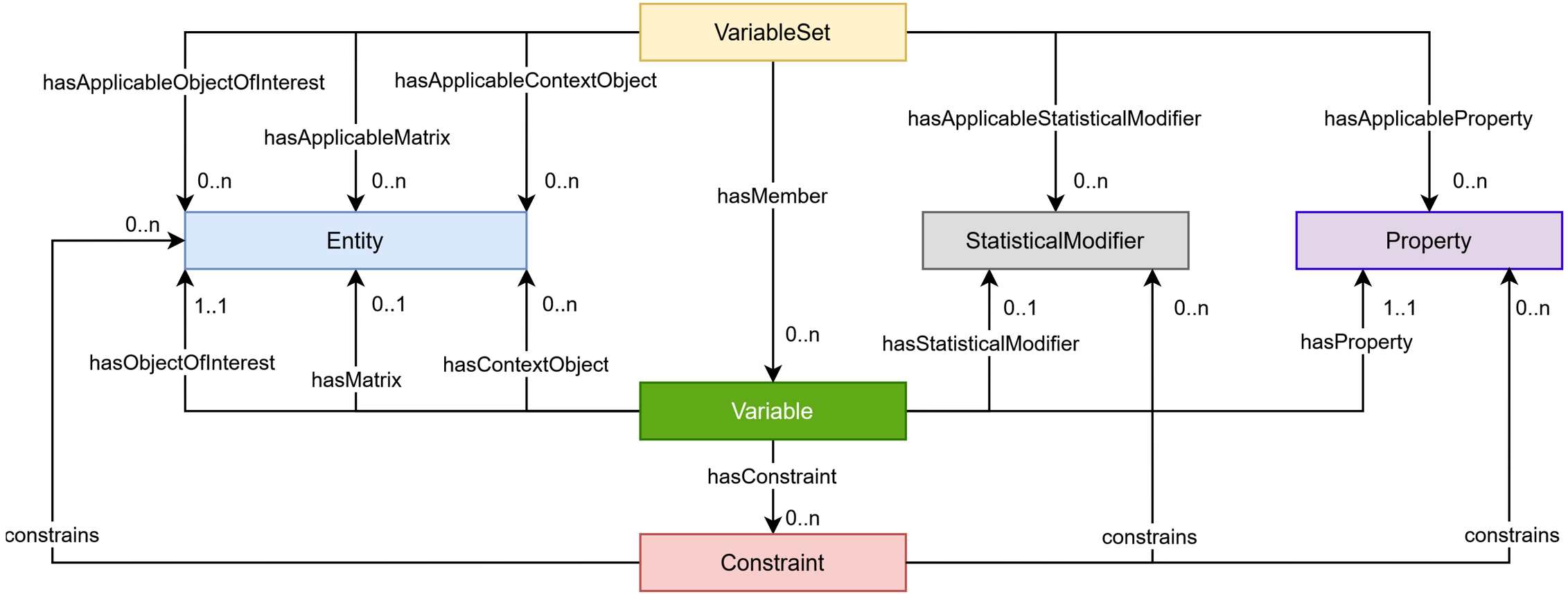


I-ADOPT core ontology



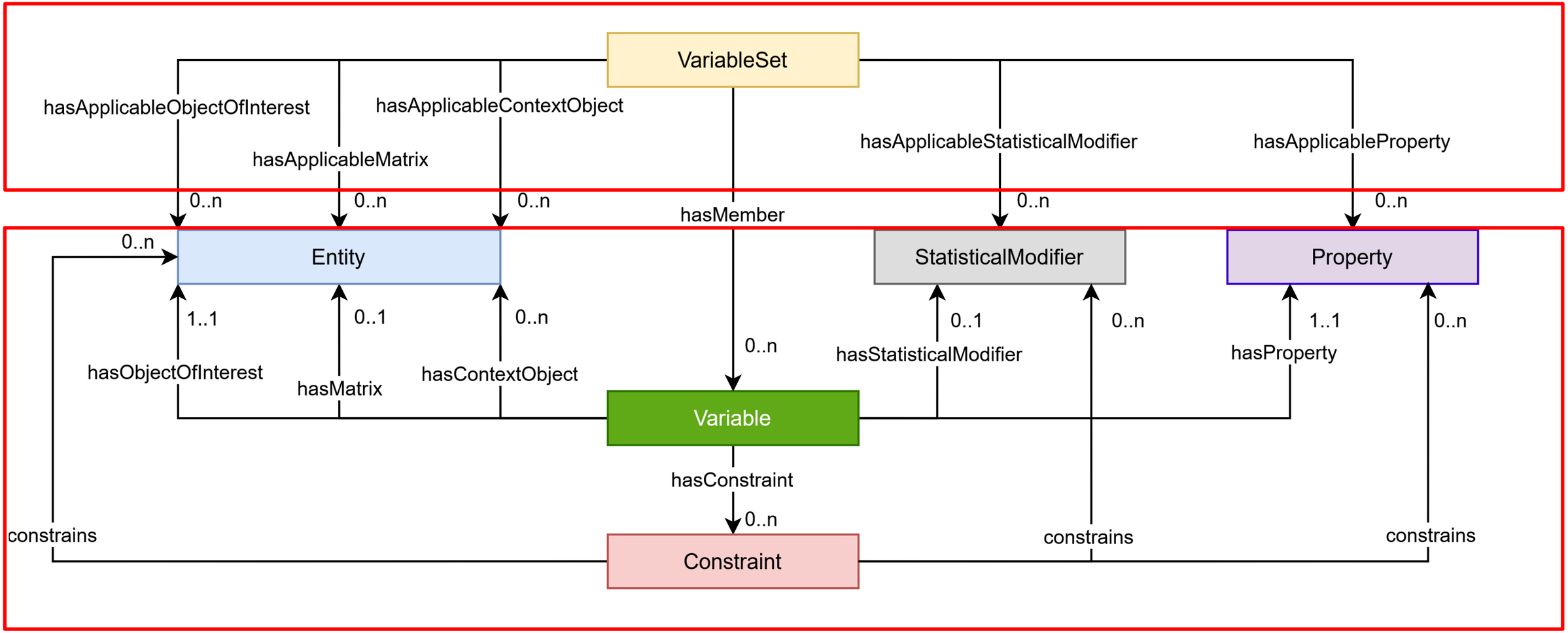
The I-ADOPT Ontology – Variableset vs Variable

https://w3id.org/iadopt/

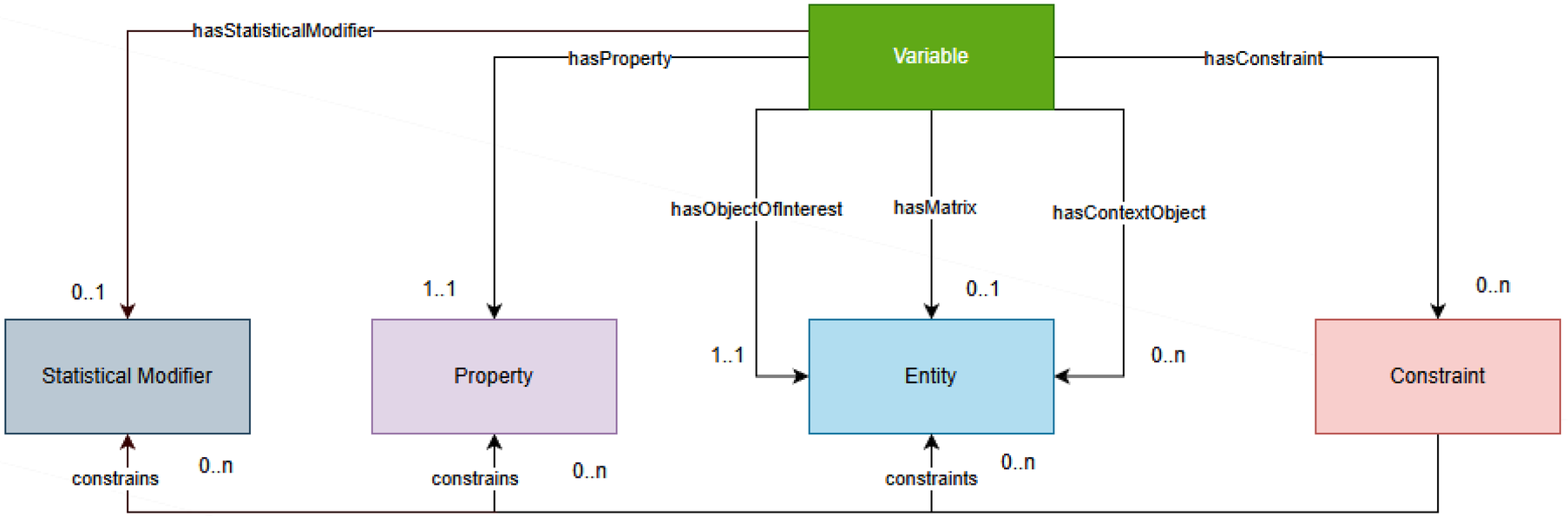


The I-ADOPT Ontology – Variableset vs Variable

https://w3id.org/iadopt/

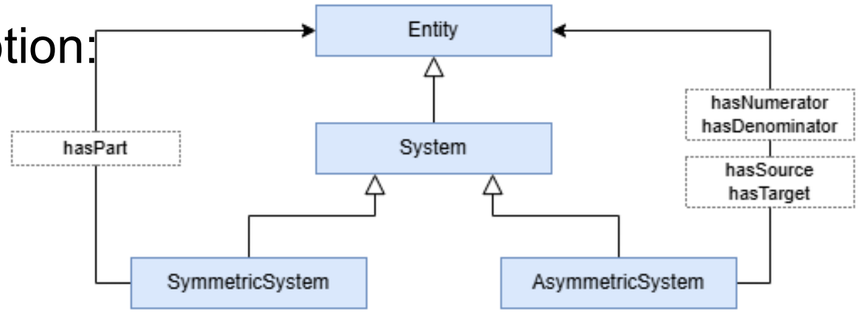


The I-ADOPT Core Ontology



Entities can have different roles in the variable description:

- Object of Interest (1..1)
- Matrix (0..1)
- Context Objects (0..n)



Entities can be represented as simple concepts or a system of
as systems of entities

Radial velocity (away from) relative to instrument standard deviation over the coverage period water current at the sea surface by high frequency radar

1. Split sentence in atomic components
2. Omit I-ADOPT irrelevant parts
3. Assign description roles
4. Link to domain vocabulary concepts
5. Visualize the model
6. Express it with a knowledge representation language
7. Publish it (e.g. as nanopublication)

Radial velocity | (away from) | relative to instrument | standard deviation | over the coverage period | water current | at the sea surface | by high frequency radar

✓ Split sentence in atomic components

Radial velocity | ~~(away from)~~ | relative to instrument | standard deviation | over the coverage period | water current | at the sea surface | ~~by high frequency radar~~

✓ Omit I-ADOPT irrelevant parts:

- value representations
- instruments

Radial velocity | relative to instrument | standard deviation | over the coverage period | water current | at the sea surface

✓ Assign description roles

Radial velocity | relative to instrument | standard deviation | over the coverage period | water current | at the sea surface

✓ Assign description roles

- Property

Radial velocity | relative to instrument | standard deviation | over the coverage period | water current | at the sea surface

✓ Assign description roles

- Property
- Object of Interest

Radial velocity | relative to instrument | standard deviation | over the coverage period | water current | at the sea surface

✓ Assign description roles
- Property
- Object of Interest
- Statistical Modifier

Radial velocity | relative to instrument | standard deviation | over the coverage period | water current | at the sea surface

✓ Assign description roles

- Property
- Object of Interest
- Statistical Modifier
- Constraint

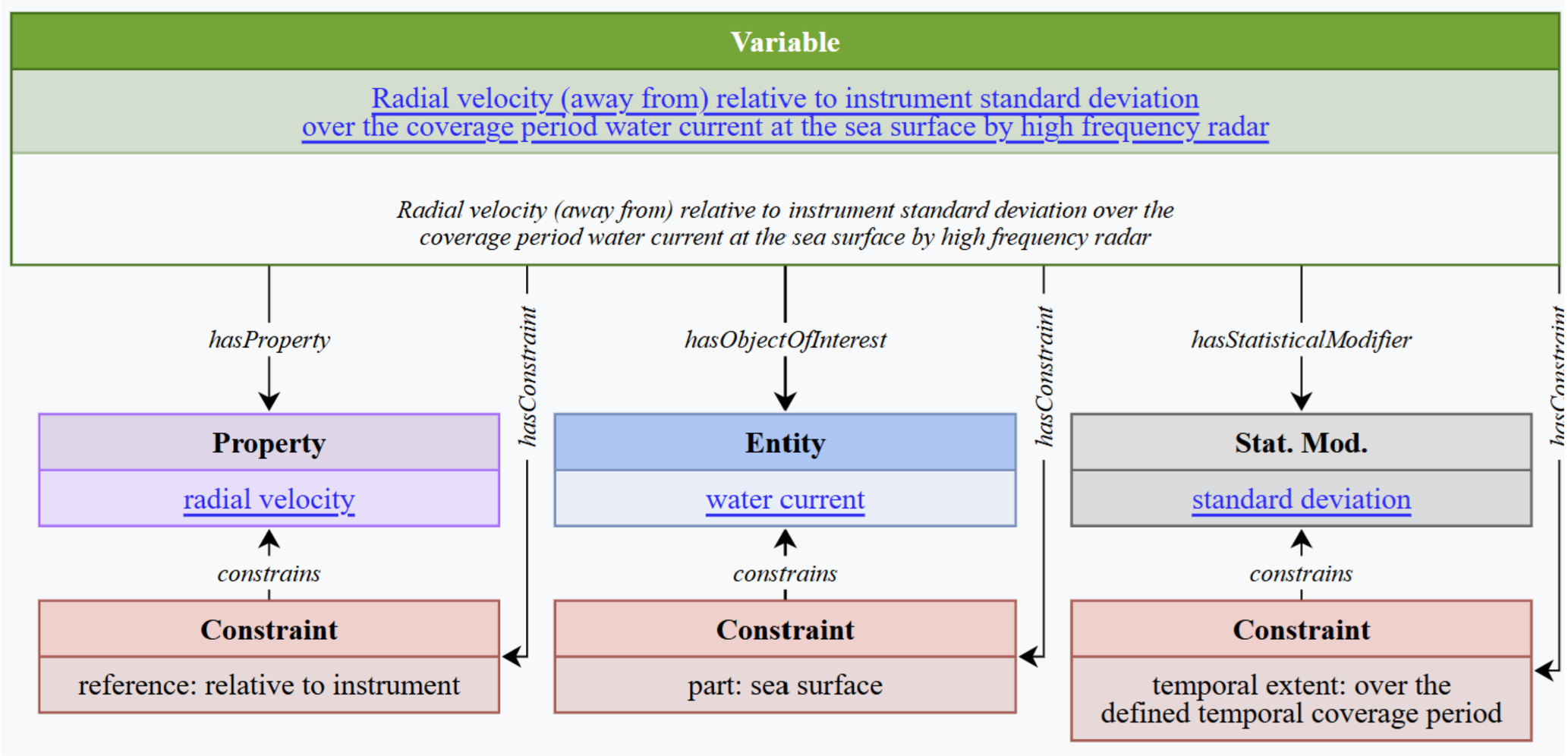
Radial velocity | relative to instrument | standard deviation | over the coverage period | water current | at the sea surface

✓ Link to domain vocabulary concepts

- Radial velocity -> S06:S0600265
- Water current -> S18:S1800022
- Standard deviation -> SIO:SIO_000770

Radial velocity | relative to instrument | standard deviation | over the coverage period | water current | at the sea surface

✓ Model and visualize the variable



https://sirkos.github.io/iadopt-vis/

Radial velocity (away from) relative to instrument standard deviation over the coverage period water current at the sea surface by high frequency radar

```
@prefix iop: <https://w3id.org/iadopt/ont/> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix skos: <http://www.w3.org/2004/02/skos/core#> .

<http://example.org/iadopt/RDVAST>
   a
      iop:Variable ;
   rdfs:label
      "Radial velocity (away from) relative to instrument standard deviation over the coverage
period water current at the sea surface by high frequency radar" ;
   rdfs:comment
      """Radial velocity (away from) relative to instrument standard deviation over the coverage
period water current at the sea surface by high frequency radar
""" ;
   iop:hasObjectOfInterest
      <http://vocab.nerc.ac.uk/collection/S18/current/S1800022/> ;
   iop:hasStatisticalModifier
      <http://semanticscience.org/resource/SIO_000770> ;
   iop:hasProperty
      <https://vocab.nerc.ac.uk/collection/S06/current/S0600265/> ;
   iop:hasConstraint
      [ a iop:Constraint ;
         rdfs:label "reference: relative to instrument" ;
         iop:constrains <https://vocab.nerc.ac.uk/collection/S06/current/S0600265/> ;
      ] ,
      [ a iop:Constraint ;
         rdfs:label "temporal extent: over the defined temporal coverage period" ;
         iop:constrains <http://semanticscience.org/resource/SIO_000770> ;
      ] ,
      [ a iop:Constraint ;
         rdfs:label "part: sea surface" ;
         iop:constrains <http://vocab.nerc.ac.uk/collection/S18/current/S1800022/> ;
      ] .

<http://vocab.nerc.ac.uk/collection/S18/current/S1800022/>
   a
      iop:Entity ;
   rdfs:label
      "water current" .

<http://semanticscience.org/resource/SIO_000770>
   a
      iop:StatisticalModifier ;
   rdfs:label
      "standard deviation" .

<https://vocab.nerc.ac.uk/collection/S06/current/S0600265/>
   a
      iop:Property ;
   rdfs:label
      "radial velocity" .
```

✓ Represent as RDF Turtle file

I-ADOPT Design Guide

I-ADOPT Design Guide

Workflow:

1. Use the I-ADOPT Visualizer to compose your variable.
2. Follow Decomposition Steps
3. Find appropriate concepts and add the IRIs in the visualizer
4. Copy RDF and safe it in the I-ADOPT Examples Repository
5. Create an Issue in the I-ADOPT Examples Repository, copy the graphic into the issue and use the labels to describe with patterns you used

1. Step:

Analyse the sentence and, if necessary, rewrite it to provide a clear interpretation of a phenomenon or entity. The help of a domain expert might be useful. Leave out details about the unit of measurements, the method and the instrument but keep the rest.

E.g. Atmospheric optical thickness of particulate organic matter at 550nm under ambient conditions.

-> Optical thickness of organic particle in aerosol in atmosphere at 550nm (ambient conditions is already captured in the term atmosphere).

2. Step:

Segment the text into atomic parts based on meaning rather than syntax. Ensure that composite terms remain intact if further decomposition would result in a loss of specialized context. Example: "atmospheric boundary layer".

Use one PATTERN for each description component (apart from constraints for which more might be needed), guided by decision trees

3. Step

Determine the Property

Be as precise as possible and use a known physical quantity rather than a community convention to increase the clarity and reusability of the terms. In case the description uses a community convention use the physical quantity and add a *constraint: type* on it.

Decision trees:

- qualitative properties
- quantitative properties
- derived properties

I-ADOPT Variable Design Patterns

To ensure that variables are modeled consistently we need implementation blueprints so called Variable Design Patterns: reusable, structured modeling solutions

A Pattern:

- ❑Applies to one or more description components
- ❑Defines if constrains play a role
- ❑Can have associated patterns
- ❑Can have associated mapping rules

```
type: Variable Design Pattern
id: https://github.com/mablue/I-ADOPT-patterns-playground/blob/main/pattern/VDP16.yaml
name: Property - length
comment: Length is a fundamental quantity.
design: Length requires a reference point, a constraint: datum.
instances:
     - width
     - depth
     - height
involved_components:
    - Property
    - Constraint (datum)
  constrained:
    - Property
associated_pattern:
https://github.com/mablue/I-ADOPT-patterns-playground/blob/main/pattern/VDP68.yaml
associated_mapping: ~
example: height of atmospheric boundary layer
gh_issue: https://github.com/mablue/I-ADOPT-examples-playground/issues/10
```

I-ADOPT LLM-enabled Service based on design patterns

with the human in the loop in development

User Prompt

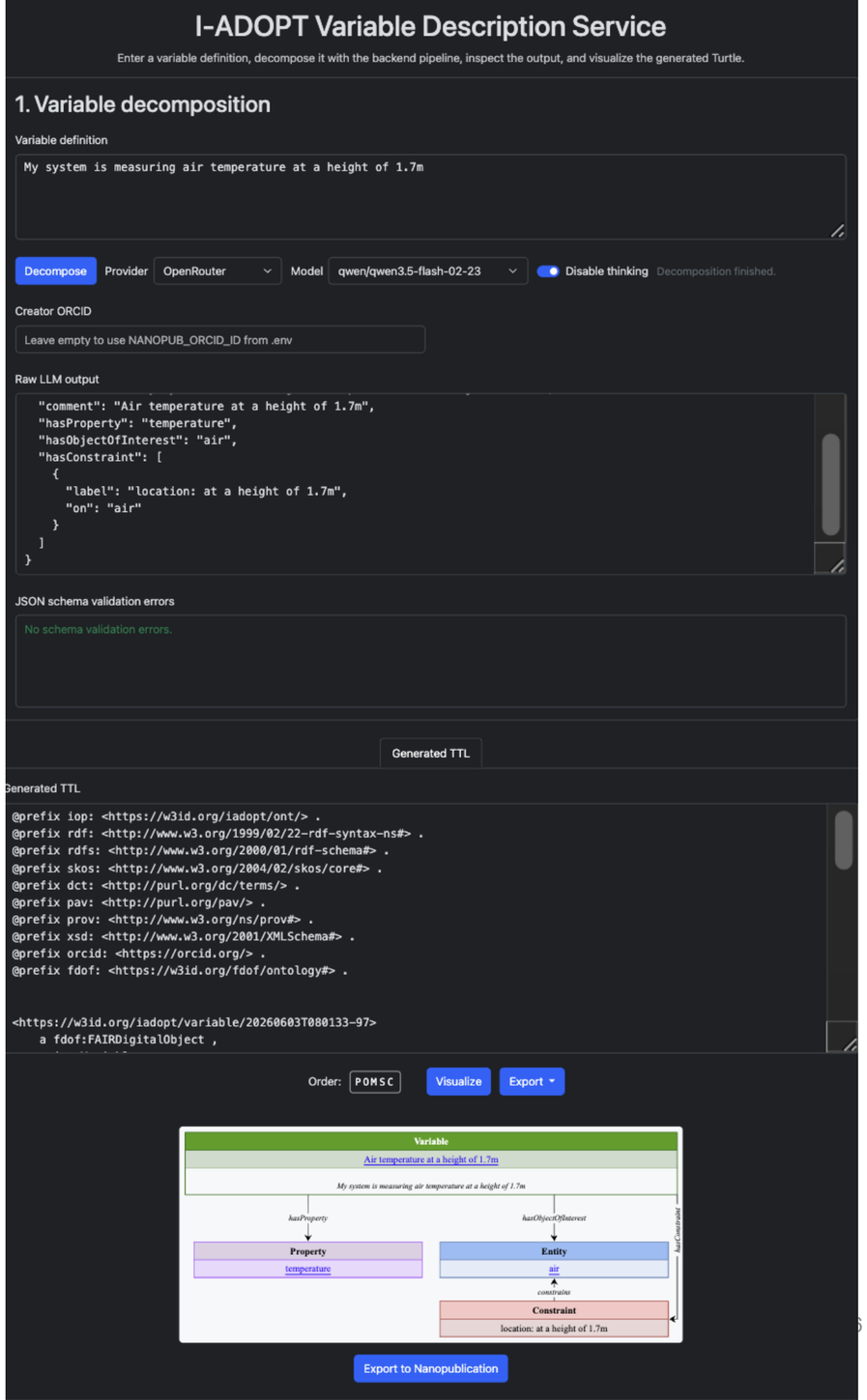
- ❑Extract
- ❑Find
- ❑Atomize
- ❑Assign roles
- ❑Annotate
- ❑Visualize
- ❑Represent
- ❑Publish
- ❑Map

EOSC | FAIR2Adapt

NFDI4Earth

CDIF 4 | EOSC

I-ADOPT service



Thank you!



Visit us at https://i-adopt.github.io/